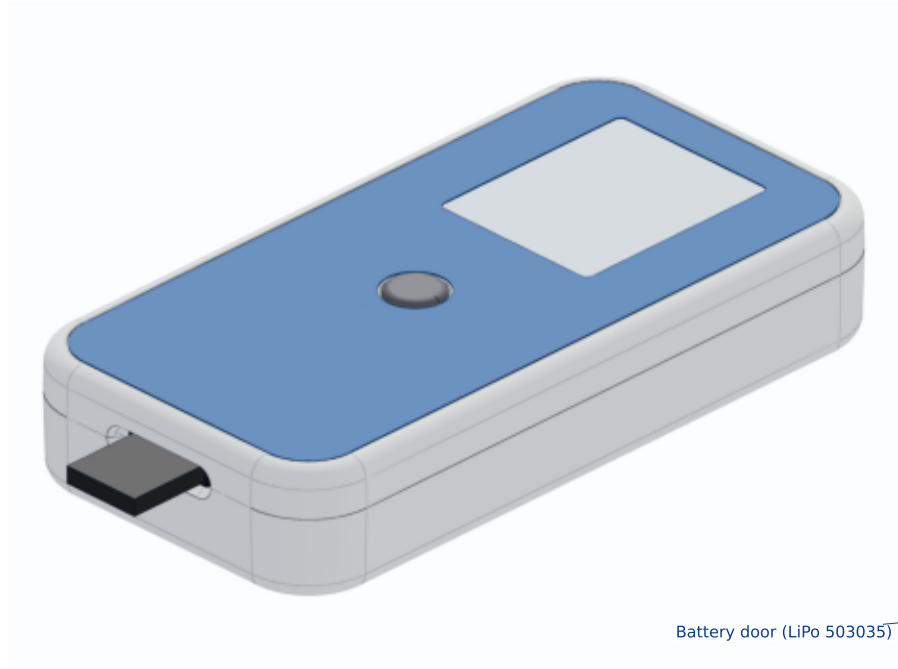
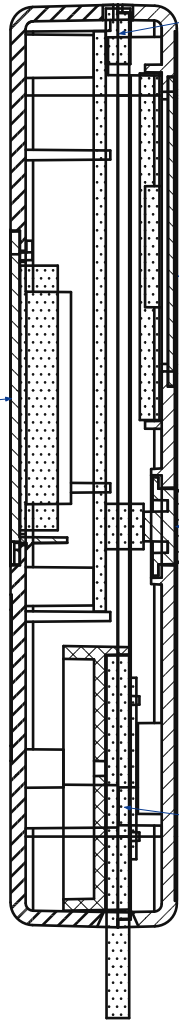


ISO VIEW (NTS)



SECTION A-A (x = 0) 1:1

ON A-A (x = 0) 1:



USB-C (split on P/L)

PMMA lens 0.8 + PC overlay

Button cap on 6x6x5 tact

Cuvette groove in black optical block

BILL OF MATERIALS

#	Part no.	Description	Material	Qty	Process / tool
1	HS-101	Front housing	ABS white	1	Inj. mould - Tool A (cav 1)
2	HS-102	Rear housing	ABS white	1	Inj. mould - Tool A (cav 2)
3	HS-103	Battery door	ABS white	1	Inj. mould - Tool A (cav 3)
4	HS-104	Power button cap	ABS black	1	Inj. mould - Tool B (cav 1)
5	HS-105	Optical block (light-tight)	ABS black	1	Inj. mould - Tool B (cav 2)
6	HS-201	Display lens 33.8x40.8x0.8	PMMA, HC one side	1	Laser cut + 3M 9471LE
7	HS-202	Graphic overlay 54.6x114.6	PC 0.175 + 7952MP	1	Screen print + die cut
8	EJOT PT K25x16	Housing screw	Steel Zn	4	Torque 0.35 Nm
9	EJOT PT K25x6	PCB + optical block screw	Steel Zn	4	Torque 0.25 Nm
10	3M SJ5302	Bumpon foot dia 7.9	PU	4	-
11	(PCB)	Main PCB 76.75x49.94x1.6	FR4	1	thk 1.6 +/-0.10 (spec!)
12	(bought)	1.8in TFT ST7735, LiPo 503035, 6x6x5 tact	-	1 ea	per electronics BOM

NOTES

1. Datum A = parting plane (P/L), z = 14.2 from the rear skin. B = centre line X. C = centre line Y.
2. All plastics: ABS, mould shrink 0.5 % built into the tool STEP data (see HS-300 sheets).
3. Parting plane runs through the USB-C centre and cuvette mouth: NO slides / lifters in any tool.
4. Housing closure: rim lap joint + 4 x PT K25x16 from the rear. Top two screws pass through the PCB top holes (PCB NOT clamped: 0.30 hold-down gap). Bottom PCB holes: 2 x PT K25x6.
5. White ABS is translucent: the optical path is enclosed in the BLACK optical block (HS-105).
6. Tolerances: general per sheet 6; critical fits and 15 stack-ups on sheet 6.
7. Colour variants (teal / grey / purple ...) come from the printed overlay only - one set of tools.
8. OPEN ITEMS: confirm PCB component heights (≤ 3.0 below, TFT FPC route), USB-C overhang 3.0 mm past PCB edge, tact switch at (0, -8), cuvette 14.0 x 3.0 x L, LiPo 503035 wiring.

LEV RAM LIFESCIENCES

HemoSure Hb meter casing

GENERAL ARRANGEMENT, STACK-UP & BILL OF MATERIALS

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART HemoSure casing assembly	DWG NO. HS-100	REV A
MATERIAL see BOM	SCALE 1:1 / NTS	SHEET 1 / 10
FINISH -	UNITS mm	PROJ. 1st angle (ISO E)
MASS 42 g (plastics)	DATE 2026-09-28	STATUS DRAFT - DFM review

Feature	Nominal	Tol.	Steel-safe
Boss pattern (both halves)	43.32 x 70.13	+0.05 / -0.05	-
Outer profile at P/L	62.00 x 122.00	+0.10 / -0.10	-
PCB seat below P/L	3.20	+0.05 / -0.05	leave long
Tongue thickness x height	0.90 x 1.50	0 / +0.05 / -0.05 / -0.05	cut thick
USB-C opening (split on P/L)	9.60 x 4.00	+0.10 / 0	cut small
Door recess	37.30 x 44.30	+0.10 / 0	cut small
Door recess depth	1.25	+0.05 / -0.05	cut shallow
Screw clearance / c'bore	dia 2.9 / dia 5.3	+0.10 / 0	-
PCB crush-rib tip to PCB edge	0.05	+0.05 / -0.05	cut tight
Optical-block boss top below P/L	7.20	+0.05 / -0.05	leave long

LEV RAM LIFESCIENCES REAR HOUSING		HemoSure Hb meter casing Drawn: Claude (AI) for S. Saboo Chk: Appr:
PART	DWG NO.	REV
Rear housing	HS-102	A
MATERIAL	SCALE	SHEET
ABS natural white (LG HI-121H / PA-757 eq.)	1:1	3 / 10
FINISH	UNITS	PROJ.
Out VDI 3400 Ref 24 / In SPI B-2	mm	1st angle (ISO E)
MASS	DATE	STATUS
22.3 g	2026-09-28	DRAFT - DFM review

A



C



HS-103 SECTION (x = 0)



HS-104 BUTTON - section 4:1



NOTES

HS-103 door: ABS white, same resin lot as HS-102.
 Flush 0/0.15 to rear skin; gap 0.15-0.25/side.
 HS-104 button: ABS black (or TPU 90A if softer
 feel wanted - same tool). Pad-print power icon.
 HS-105 optical black: ABS BLACK, opaque at 2 mm
 (transmission < 0.1 % @ 400-700 nm - verify).
 Groove Ra <= 0.8 (SPI B-1) to avoid scratching
 the cuvette optical window. Parting at underside.
 Add 0.5 x 45° lead-in at groove entry (T1).
 All: draft 1° modelled on every wall.

Cap rests on the tact actuator; 0.45 play to the housing face (stack 3).

Mould windows under hooks/bump let the cavity form the hook undersides (no lifters).

HS-105 SECTION D-D (y = -40, optical axis) 2.5:1



LEVRAM LIFESCIENCES		HemoSure Hb meter casing	
BATTERY DOOR / BUTTON CAP / OPTICAL BLOCK		Drawn: Claude (AI) for S. Saboo Chk: ____ Appr: ____	
PART Small moulded parts	DWG NO. HS-103/104/105	REV A	
MATERIAL 103: ABS white 104, 105: ABS black	SCALE 2:1	SHEET 4 / 10	
FINISH -	UNITS mm	PROJ. 1st angle (ISO E)	
MASS 2.1 g / 0.3 g / 2.1 g	DATE 2026-09-28	STATUS DRAFT - DFM review	

A

B

C

D

E

F



A

B

C

D

E

F



A

B

C

D

E

F

A

B

C

D

E

F

A

B

C

D

E

F

A

B

C

D

E

F

A

B

C

D

E

F

GENERAL TOLERANCES - ABS (unless stated)

Size range (mm)	Tool-bound (single half)	Across P/L / not tool-bound
0 - 6	+/-0.10	+/-0.15
> 6 - 30	+/-0.15	+/-0.20
> 30 - 120	+/-0.25	+/-0.35
> 120 - 315	+/-0.40	+/-0.55

BASIS

Basis: ISO 20457 tolerance groups for ABS (shrink 0.4-0.7 %), normal production.
 Angles $\pm 0.5^\circ$. Radii ± 0.2 . Flatness P/L 0.10. Position of holes ± 0.05 where on the PCB / screw pattern, else general. Critical dims are T1-tuned ('steel-safe').
 Steel-safe = cut the tool so that metal can only be REMOVED to reach nominal:
 holes/pockets cut small, bosses/pins left long, recesses cut shallow.

CRITICAL / FUNCTIONAL DIMENSIONS

Part	Feature	Nominal	+	-
Front	Outer profile at P/L	62.00 x 122.00	+0.10	-0.10
Front	Display window (at pocket floor)	30.00 x 37.00	+0.10	0
Front	Lens pocket	34.00 x 41.00	+0.10	0
Front	Lens pocket depth	0.95	+0.05	-0.05
Front	Overlay recess	55.00 x 115.00	+0.10	0
Front	Overlay recess depth	0.25	+0.03	-0.03
Front	Button hole	dia 10.30	+0.08	0
Front	Screw pilot (PT K25)	dia 2.00	+0.05	0
Front	Top boss end below P/L	1.30	+0.05	-0.05
Front	Bottom boss end above P/L	0.10	+0.05	-0.05
Front	Lap step width x depth	1.00 x 1.70	+0.05	0
Front/Rear	Boss pattern (both halves)	43.32 x 70.13	+0.05	-0.05
Rear	Outer profile at P/L	62.00 x 122.00	+0.10	-0.10
Rear	PCB seat below P/L	3.20	+0.05	-0.05
Rear	Tongue thickness x height	0.90 x 1.50	0 / +0.05	-0.05 / -0.05
Rear	USB-C opening (split on P/L)	9.60 x 4.00	+0.10	0
Rear	Door recess	37.30 x 44.30	+0.10	0
Rear	Door recess depth	1.25	+0.05	-0.05
Rear	Screw clearance / c'bore	dia 2.9 / dia 5.3	+0.10	0
Rear	PCB crush-rib tip to PCB edge	0.05	+0.05	-0.05
Rear	Optical-block boss top below P/L	7.20	+0.05	-0.05
Door	Plate outline x thickness	37.00 x 44.00 x 1.20	0	-0.10 / -0.05
Door	Latch arm t x bump reach	0.80 x 0.80	+0.03 / +0.05	-0.03 / -0.05
Button	Cap dia / flange dia	10.00 / 13.00	0 / +0.05	-0.05 / -0.05
Button	Stem tip to flange top	1.95	+0.05	-0.05
Optical	Cuvette groove width x depth	14.30 x 3.30	+0.05	0
Optical	Floor under groove / aperture	1.50 / dia 2.00	+/-0.05	+/-0.03
Optical	Ear hole pitch	24.00	+0.05	-0.05

SUPPLIER-SIDE TOLERANCES

BOUGHT-IN SPECS THE STACKS DEPEND ON (put on the POS):

PCB: 1.60 $\pm$ 0.10 (not IPC default $\pm$ 10 %), outline $\pm$ 0.10 routed,
holes dia 3.20 $\pm$ 0.05, hole pattern $\pm$ 0.05, USB-C overhang 3.0 $\pm$ 0.1.
Tact switch 6x6, height 5.00 $\pm$ 0.10, travel 0.25, 1.6 N.
USB-C receptacle 8.94 x 3.26 shell, top-mount, centre 1.63 above PCB.
TFT 1.8in ST7735: glass 34.00 x 45.83 $\pm$ 0.10, active area offset
2.0 towards the FPC-free end (confirm module drawing).
Cuvette 14.00 $\pm$ 0.05 x 3.00 $\pm$ 0.05 (confirm drawing).

LATCH STRAIN (HS-103): $e = 1.5 \text{ t d} / L^2$, $t \geq 0.8$, $L \geq 6.3$:
nominal $d \geq 0.5 \rightarrow 1.5 \%$; max $d \geq 0.75 \rightarrow 2.3 \%$ (ABS limit $\sim 2.5 \%$).
SCREW BOSSES: PT K25 in ABS, pilot 2.0, engagement ≥ 5.0 (2d).

STACK-UP ANALYSIS (gaps in mm; + = clearance)

#	Stack	Nom.	Worst case	RSS	Accept	WC	RSS
1	Housing closure: front vs rear bottom screw bosses	+0.100	+0.00 .. +0.20	+0.03 .. +0.17	+0.00 .. +0.25	OK	OK
2	PCB hold-down gap (front top boss to PCB top)	+0.300	+0.04 .. +0.56	+0.13 .. +0.47	+0.00 .. +0.60	OK	OK
3	Button: flange-to-face play with cap resting on the switch	+0.450	+0.10 .. +0.80	+0.28 .. +0.62	+0.05 .. +0.90	OK	OK
4	Button protrusion above the front face at rest	+0.600	+0.15 .. +1.05	+0.20 .. +1.00	+0.15 .. +1.10	OK	OK
5	Lens below overlay floor (flush)	+0.100	-0.01 .. +0.21	+0.03 .. +0.17	-0.02 .. +0.25	OK	OK
6	Overlay flush in its recess	+0.025	-0.03 .. +0.08	-0.01 .. +0.06	-0.03 .. +0.10	OK	OK
7	Battery door flush with the rear skin	+0.050	-0.05 .. +0.15	-0.02 .. +0.12	-0.05 .. +0.20	OK	OK
8	Battery door gap per side	+0.150	+0.05 .. +0.25	+0.08 .. +0.22	+0.05 .. +0.30	OK	OK
9	Door latch engagement over the ledge	+0.500	+0.20 .. +0.80	+0.33 .. +0.67	+0.20 .. +0.80	OK	OK
10	Lap joint radial clearance (tongue to front step)	+0.100	-0.01 .. +0.21	+0.03 .. +0.17	-0.02 .. +0.25	OK	OK
11	Lap joint: tongue height vs step depth	+0.200	+0.10 .. +0.30	+0.13 .. +0.27	+0.05 .. +0.35	OK	OK
12	USB-C: vertical clearance in the opening (worst side)	+0.370	+0.09 .. +0.65	+0.16 .. +0.58	+0.00 ..	OK	OK
13	USB-C: lateral clearance in the opening (worst side)	+0.330	+0.00 .. +0.66	+0.14 .. +0.52	+0.00 ..	OK	OK
14	PCB hole vs M2.5 housing screw (top holes)	+0.350	+0.00 .. +0.70	+0.16 .. +0.54	+0.00 ..	OK	OK
15	Cuvette lateral clearance in the groove (per side)	+0.150	+0.10 .. +0.20	+0.11 .. +0.19	+0.05 .. +0.25	OK	OK

CONTRIBUTORS

- front boss end above P/L +0.10 ±0.05; rear boss top at P/L +0.00 ±0.05
Gap must stay ≥ 0 so the perimeter lap closes first (no visible P/L gap).
2. rear standoff seat below P/L +3.20 ±0.05; PCB thickness -1.60 ±0.16; front top boss end below P/L -1.30 ±0.05
Boss must never crush the PCB; PCB taken at IPC +/-10 % here.
3. front inner face above P/L +5.80 ±0.05; rear standoff seat below P/L +3.20 ±0.05; PCB thickness -1.60 ±0.10; tact switch height -5.00 ±0.10; cap: stem tip to flange top -1.10 ±0.10
Play $> 0 \Rightarrow$ the housing can never pre-press the switch.
4. cap: flange top to cap top +3.05 ±0.05; flange-top position (= face - play) -2.45 ±0.40
Stays proud of the overlay for a tactile press.
5. lens pocket depth +0.95 ±0.05; lens thickness (PMMA) -0.80 ±0.05; lens tape (3M 9471LE 0.05) -0.05 ±0.01
Lens must not push the overlay up (max 0.02 proud accepted).
6. recess depth +0.25 ±0.03; overlay 0.175 PC + 0.05 adhesive -0.23 ±0.03
Overlay edge must not stand proud (lift / peel).
7. door recess depth +1.25 ±0.05; door plate thickness -1.20 ±0.05
0.05 proud to 0.15 sunk accepted.
8. recess width / 2 +18.65 ±0.05; door width / 2 -18.50 ±0.05
Visible gap; door must drop in freely.
9. bump reach from arm face +0.80 ±0.05; arm-to-edge clearance -0.30 ±0.05; door float in recess +0.00 ±0.15; opening edge position +0.00 ±0.05
Engagement > 0.2 ; max deflection sets the latch strain (see note).
10. step face offset (front) +1.00 ±0.03; tongue outer offset (rear) -1.10 ±0.03; front/rear profile mismatch +0.00 ±0.05
Tongue 0.90 0/-0.05 makes the real minimum 0.02 (+0.05 thinner tongue); zero-clearance local touch is acceptable.
11. step depth (front) +1.70 ±0.05; tongue height (rear) -1.50 ±0.05
Tongue must never bottom out.
12. half opening height +2.00 ±0.05; half receptacle height -1.63 ±0.03; receptacle centre vs P/L (seat, PCB, part) +0.00 ±0.20
Plug must enter without touching the housing.
13. half opening width +4.80 ±0.05; half receptacle width -4.47 ±0.03; PCB lateral float (crush-rib clearance + outline) +0.00 ±0.15; receptacle placement on PCB +0.00 ±0.10
14. (hole - screw) / 2 +0.35 ±0.05; PCB float +0.00 ±0.15; hole position on PCB +0.00 ±0.10; boss position +0.00 ±0.05
Screw must pass without loading the PCB sideways.
15. groove width / 2 +7.15 ±0.03; cuvette width / 2 -7.00 ±0.03
Confirm the cuvette drawing: this assumes 14.00 +/-0.05.

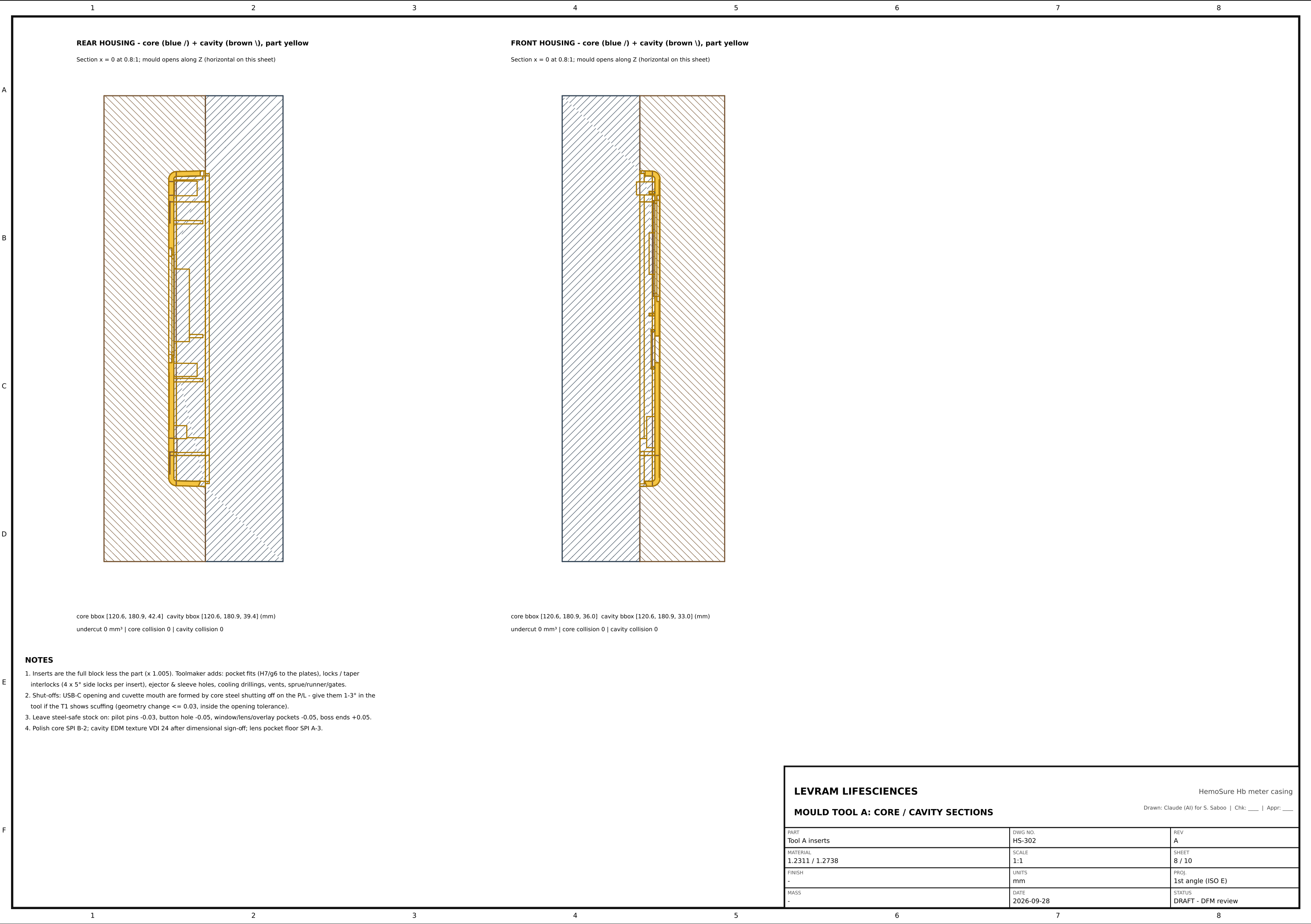
LEV RAM LIFESCIENCES

TOLERANCES, FITS & STACK-UP ANALYSIS

HemoSure Hb meter casing

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART All moulded parts	DWG NO. HS-110	REV A
MATERIAL ABS	SCALE -	SHEET 6 / 10
FINISH -	UNITS mm	PROJ. 1st angle (ISO E)
MASS -	DATE 2026-09-28	STATUS DRAFT - DFM review



2

3

4

5

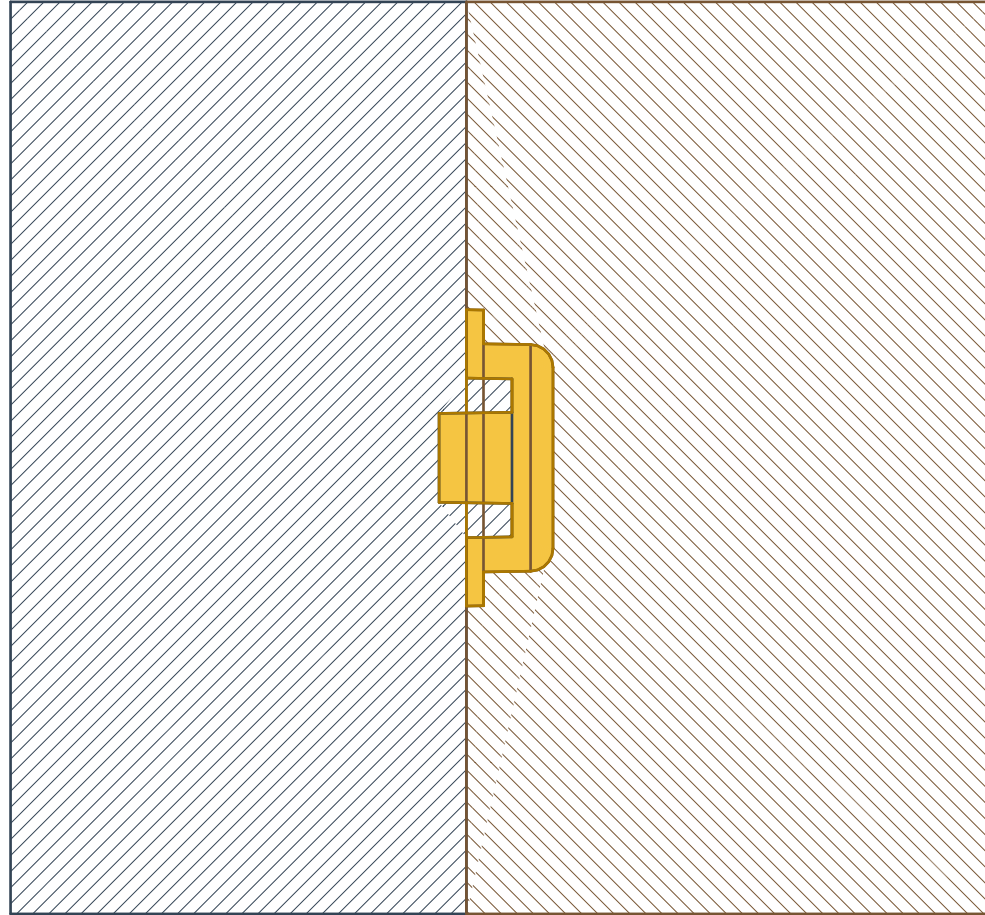
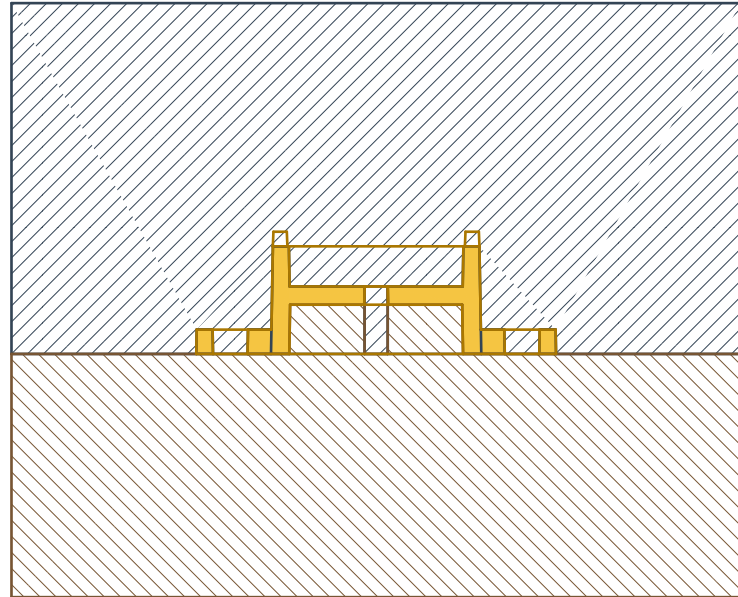
6

7

8

OPTICAL BLOCK - core / cavity section (1.6:1)

BUTTON CAP - core / cavity section (3.0:1)



undercut 0.0002 mm³ | collisions 0.0002 / 0 mm³

undercut 0 mm³ | collisions 0 / 0 mm³

NOTES

1. Both parts black -> one family tool; if the button must be another colour, split to its own 1-cavity tool.
2. Optical block groove is the optical-path datum: hold $14.30 \pm 0.05/0$ and floor 1.50 ± 0.05 .
3. Heat-stake pins dia 1.2: vent the pin tips (0.01) to avoid short shots.
4. Verify opacity on T1 parts: no visible light through 2 mm with a torch in a dark room; spectrometer $< 0.1\%$.

Item	Tool B data
Cavities	1 x button + 1 x optical block, black ABS
Parting	Button: flange underside. Block: underside
Shrink	0.50 % (in insert STEP)
Feed	Sub gate dia 0.8 (button, on stem side), sub gate dia 1.0 (block, on ear underside)
Ejection	Button: sleeve on stem; block: 4 pins on ears
	+ 2 pins on the end walls
Polish	Groove + floor SPI B-1 (cuvette contact)
Proj. area	~7.5 cm² -> 30-50 t press
Shot	2.4 g parts + ~2 g runner

LEVRAM LIFESCIENCES

HemoSure Hb meter casing

MOULD TOOL B (1+1, BLACK ABS)

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART Tool B: button + optical block	DWG NO. HS-303	REV A
MATERIAL Inserts 1.2311 P20 / cavity 1.2083 for polish	SCALE 2:1	SHEET 9 / 10
FINISH -	UNITS mm	PROJ. 1st angle (ISO E)
MASS -	DATE 2026-09-28	STATUS DRAFT - DFM review

AUTOMATED CAD CHECKS (this revision)

Part	Vol. cm³	Mass g	Undercut mm³	Core coll.	Cav. coll.	Faces < 0.2° draft (>1 mm tall)	Parting z
Rear Housing	21.25	22.3	0	0	0	0	14.2
Front Housing	14.28	15.0	0	0	0	0	14.2
Battery Door	1.97	2.1	0	0	0	0	1.2
Button Cap	0.29	0.3	0	0	0	0	18.8
Optical Block	2.01	2.1	0.0002	0.0002	0	0	7.0

Assembly interference check, all part pairs incl. PCB/TFT/USB-C/tact/LiPo/cuvette/LED board: NONE

METHOD

- How the checks work (hemosure_casing.py + mold_tools.py, re-run on every change):
- Undercut: steel that is blocked from BOTH the core and the cavity direction (sweep of the part along +Z and -Z).
 - Collision: core insert intersected with the part swept along the core opening direction (and same for the cavity).
 - Draft: every face's angle to the draw axis; faces taller than 1 mm with < 0.2° are listed (target: none).
 - Stack-ups: 15 worst-case + RSS stacks, sheet 6. All pass worst case at this revision.

DESIGN RULES APPLIED

Wall 2.00 nominal (ABS range 1.2-3.5); thinnest: lens-pocket ledge 0.80, door ledge 0.75, lap skirt 1.00.
Ribs 1.0-1.2 (50-60 % of wall), rib height <= 3x wall except PCB posts (gusseted, sleeve-ejected).
Boss OD 5.0-5.6 / hole 2.0-2.9: boss wall 1.35-1.5 (<= 0.75 x wall) -> no sink on the rear skin.
All inside corners R >= 0.5 at wall/floor (R1.0 modelled at the main wall-floor junction).
Flow length / wall (L/t) max ~42 from a single side gate - well inside ABS limit (~150).
Critical cosmetic: front overlay field, rear skin, P/L line. Non-cosmetic: interior, optical block.

VALIDATION

- T0/T1 TRIAL CHECKLIST (for the tool room)
1. Dry ABS 80 °C / 3 h; dew point <= -20 °C.
 2. Short-shot study -> fill balance of the family tool.
 3. Measure the critical-dims table (sheet 6) on 5 shots/cavity;
Cpk >= 1.33 on: boss pattern, PCB seat, window, lens pocket.
 4. Assemble with a real PCB + TFT + LiPo: USB-C plug insertion, button click (no pre-press, 0.25 travel), door open/close x 50.
 5. Drop test 1.0 m x 6 faces (IEC 61010-1 / 60068-2-31 style).
 6. Light-leak test of the optical cell with the cuvette inserted.
 7. Disinfectant wipe x 500 (70 % IPA, 0.5 % NaOCl): no ESC cracks at bosses (if cracking: switch to PC/ABS, shrink 0.5-0.7 %).
 8. Then texture the cavities (VDI 24) and re-check P/L step.

MOULDING START POINT

Process window (start point): melt 230 °C, mould 55 °C, injection speed medium, hold 60-70 % of fill pressure for 4-5 s, cooling 9 s, ejection at part temp < 85 °C. Gate freeze study to set the hold time. Regrind <= 15 % on cosmetic parts.

DATA PACKAGE

Files (all regenerated from cad/hemosure_casing.py - change a parameter, re-run, everything updates):
cad/HemoSure_*.step (parts + assembly) mold/HemoSure*_{CORE,CAVITY}.step (steel size) drawings/*.pdf
cad/tolerance_stack.py (stacks) cad/dfm_checks.py (draft) mold/mold_check_report.json (undercut / collision)

LEVRAM LIFESCIENCES

HemoSure Hb meter casing

DRAFT VERIFICATION & MOULDING SPECIFICATION

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART	DWG NO.	REV
All moulded parts	HS-310	A
MATERIAL	SCALE	SHEET
ABS	-	10 / 10
FINISH	UNITS	PROJ.
-	mm	1st angle (ISO E)
MASS	DATE	STATUS
-	2026-09-28	DRAFT - DFM review